Retrieval-Augmented Generation (RAG)

1. What is RAG?

Retrieval-Augmented Generation combines information retrieval with language generation. Instead of relying only on information encoded in model parameters, a RAG system retrieves relevant information from an external knowledge source and provides that information to the language model as context.

2. RAG Pipeline

A typical RAG pipeline contains document ingestion, text splitting, indexing, retrieval, context construction, and answer generation. Documents are first processed into searchable units. A user query is then used to retrieve relevant units, which are supplied to the LLM.

3. Retrieval

Retrieval identifies pieces of the knowledge base that are relevant to a query. Retrieval can use keyword matching, semantic similarity, or a combination of methods. The goal is to provide the generation model with useful evidence rather than the entire document collection.

4. Grounded Generation

The retrieved text is placed into the prompt or context given to the LLM. A grounded system instructs the model to base its answer on the supplied context. If the required information is absent, the application can instruct the model to state that the knowledge base does not contain the answer.

5. Benefits

RAG can make answers more traceable to source material and can allow an application to work with private or domain-specific documents without retraining the LLM. The knowledge source can also be updated independently of the language model.

6. Challenges

RAG quality depends on retrieval quality as well as generation quality. Poor chunking, irrelevant retrieval, duplicate information, missing context, and overly large context can reduce answer quality. Evaluation should therefore examine both retrieval and generated answers.